ECE 447

Lesson 34

GR2 Review Day



Schedule and Admin

- Schedule.
  - Lesson 34 - **GR2 Review Day** (today)
  - Lesson 35 - GR2 (Tuesday, 17 Nov) – Lessons 18-32, Chapters 5-8
  - Lessons 36-38 - Binary Digital System Performance, Error Correction
  - Lesson 39 - SDR Lab 8: Capstone image transfer
  - Lesson 40 - Advanced topics: OFDM, MIMO, CDMA
  - Lesson 41 - Course review
- Admin
  - **HW8**. Due tonight (Lesson 34, 13 Nov) to Gradescope.
  - **Lab 7**. Due Lesson 37 (Mon, 23 Nov) to Gradescope.
  - **HW9**. Assigned Lesson 36 (Thu, 19 Nov); due Lesson 40 (Mon, 7 Dec).

CHP 5 REVIEW: LESSONS 18-20, HWS 4 & 5

- **Sampling**
  - Impulse train in time $\leftrightarrow$ impulse train in frequency
  - Creates copies of message spectrum in frequency
  - Sinc is ideal interpolator, if $f_s \geqslant 2B$
  - Non-ideal pulses have "filtering" effect on sampled signal - equalizers reverse the filtering to obtain original signal
  - What two requirements do we have for *distortionless* filtering?

  - $G(f)e^{-j2\pi ft_0} = E(f)\tilde{G}(f) = E(f)P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$, so make equalizer transfer function

$$E(f) = \begin{cases} T_s e^{-j2\pi ft_0} \cdot |P(f)|^{-1}, & |f| \leqslant B \\ \text{Flexible}, & B < |f| < f_s - B \\ 0, & |f| \geqslant f_s - B \end{cases}$$

  - Just need to invert $P(f)$ and plug into $E(f)$! Example: 5.1-5a - find $E(f)$ if $p(t) = \Pi(\frac{t}{T_s} - \frac{1}{2})$

CHP 5 REVIEW: LESSONS 18-20, HWS 4 & 5

- **PCM**
  - Just sample, quantize, and encode
  - Message peak at $\pm m_p$, so each bin is $2m_p/L$, where $L = 2^n$ levels, $n =$ number of bits
  - Each additional bit improves the SNR how much?
  - Max data rate per bandwidth is $\frac{\text{2bps}}{\text{Hz}}$ - assuming a noiseless channel w/ bandwidth B
  - **If sampling exactly at Nyquist,** $f_s = 2B$**,** then min req'd channel bandwidth is $B_T = nB$
  - **If sampling at greater than Nyquist,** $f_s > 2B$**,** then min req'd channel bandwidth is $B_T = \frac{nf_s}{2}$ or simply the data rate nf_s divided by 2

  - Be able to solve problems like 5.2-3 or Example 5.2, where a max quantization error is given as a percentage of m_p and f_s is given as a percent of the Nyquist rate

CHP 6 REVIEW: LESSONS 22-27, HWS 6 & 7

- **Digital Communications**
  - **Line codes**
  - Be able to plot $y(t) = \sum_k a_k p(t - kT_b)$, i.e., the baseband waveform
  - Be able to use generic PSD equation in (6.11c)? Maybe, but I'd provide values for R_n
  - Band-limited vs. time-limited
  - **ISI**
  - Nyquist's First Criterion for Zero ISI:

$$p(t) = \begin{cases} 1, & t = 0 \\ 0, & t = \pm nT_b \end{cases}$$

  - Excess bandwidth f_x, roll-off factor r
  - GR will **clearly** use R_b for bit rate and R_s for symbol rate/baud
  - How do we accomplish timing synchronization?
  - Parts of eye diagrams

Chp 7 Review: Lessons 30-31, HW 8

- **Probability**
  - Bernoulli trials using bits
  - Conditional/total probability
  - Gaussian RV, Q function
  - Expected value, mean square, and variance
  - Central limit theorem

Chp 8 Review: Lesson 32, HW 8

- **Random processes**
  - An RP is an ensemble of sample functions — a random variable at every instant t
  - **Stationary**: *every* statistic is unchanged by a shift of the time origin
  - **Wide-sense stationary**: only the first two — $\overline{x(t)}$ constant and $R_x(t_1, t_2) = R_x(\tau)$
  - Which implication is the true one: stationary $\Rightarrow$ WSS, or WSS $\Rightarrow$ stationary?
  - **Ergodic**: ensemble averages equal time averages, so one sample function tells you everything
  - Autocorrelation $R_x(\tau) = \overline{x(t)x(t+\tau)}$, and PSD $S_x(f) = \mathcal{F}\{R_x(\tau)\}$
  - Why does a PSD only make sense once the process is WSS?
  - Average power, three ways to the same number: $P_x = \overline{|x|^2} = R_x(0) = \int_{-\infty}^{\infty} S_x(f)df$
  - Be ready to *name an example* of each, not just define it — that is how HW8 asked it